```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
df=pd.read_csv("/content/WA_Fn-UseC_-Telco-Customer-Churn.csv")
```

```
print("Dataset shape:", df.shape)
display(df.head())
```

```
Dataset shape: (7043, 21)
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	Phone
0	7590-VHVEG	Female	0	Yes	No	1	
1	5575-GNVDE	Male	0	No	No	34	
2	3668-QPYBK	Male	0	No	No	2	
3	7795-CFOCW	Male	0	No	No	45	
4	9237-HQITU	Female	0	No	No	2	

5 rows × 21 columns

```
print("Column names:")
print(df.columns.tolist())

print("\nDataset information:")
df.info()

print("\nMissing values:")
print(df.isnull().sum())

print("\nDuplicate rows:", df.duplicated().sum())

print("\nChurn rate:")
print(df["Churn"].value_counts())
```

```
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

Missing values:
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64

Duplicate rows: 0

Churn rate:
Churn
No     5174
Yes    1869
Name: count, dtype: int64
```

```
df = df.drop_duplicates().copy()

# Ensure charges columns are numeric
df["MonthlyCharges"] = pd.to_numeric(df["MonthlyCharges"], errors="coerce")
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")

# Drop any rows where important fields might be missing after coercion
df = df.dropna(subset=["customerID", "MonthlyCharges", "TotalCharges"])
```

```
print("Cleaned dataset shape:", df.shape)
display(df.head())
```

```
Cleaned dataset shape: (7032, 21)
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	Phone
0	7590-VHVEG	Female	0	Yes	No	1	
1	5575-GNVDE	Male	0	No	No	34	
2	3668-QPYBK	Male	0	No	No	2	
3	7795-CFOCW	Male	0	No	No	45	
4	9237-HQITU	Female	0	No	No	2	

5 rows × 21 columns

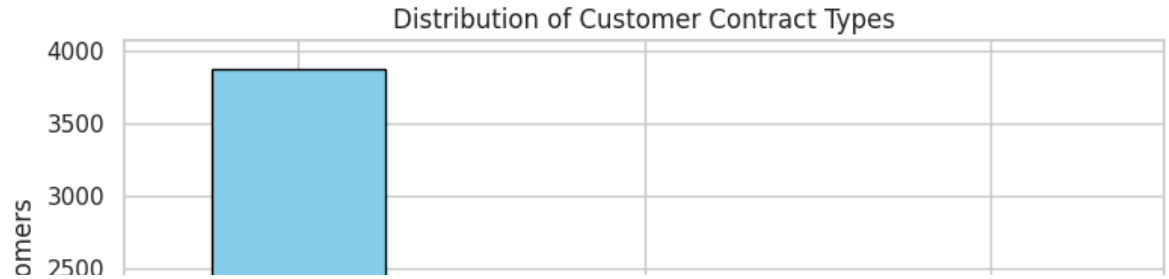
```
contract_counts = df["Contract"].value_counts()

print(contract_counts)

contract_counts.plot(
    kind="bar",
    figsize=(8, 5),
    color="skyblue",
    edgecolor="black"
)

plt.title("Distribution of Customer Contract Types")
plt.xlabel("Contract Type")
plt.ylabel("Number of Customers")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

```
Contract
Month-to-month    3875
Two year          1685
One year          1472
Name: count, dtype: int64
```




```
total_customers = df["customerID"].nunique()
churned_customers = df[df["Churn"] == "Yes"]["customerID"].nunique()

print("Total unique customers:", total_customers)
```

```
print("Customers who have churned:", churned_customers)

if total_customers > 0:
    churn_rate = (churned_customers / total_customers) * 100
    print(f"Overall churn rate: {churn_rate:.2f}%")
```

```
Total unique customers: 7032
Customers who have churned: 1869
Overall churn rate: 26.58%
```

```
total_monthly_revenue = df["MonthlyCharges"].sum()

print(f"Total monthly billing value: ${total_monthly_revenue:,.2f}")

contract_financials = (
    df.groupby("Contract")
    .agg(
        customer_count=("customerID", "size"),
        avg_monthly_charges=("MonthlyCharges", "mean"),
        total_charges=("TotalCharges", "sum")
    )
    .sort_values("total_charges", ascending=False)
)

display(contract_financials)
```

```
Total monthly billing value: $455,661.00
```

Contract	customer_count	avg_monthly_charges	total_charges
Two year	1685	60.872374	6283253.7
Month-to-month	3875	66.398490	5305861.5
One year	1472	65.079416	4467053.5

Next steps: Generate code with contract_financials | New interactive sheet

```
internet_churn = (
    df.groupby(["InternetService", "Churn"])
    .size()
    .unstack(fill_value=0)
)

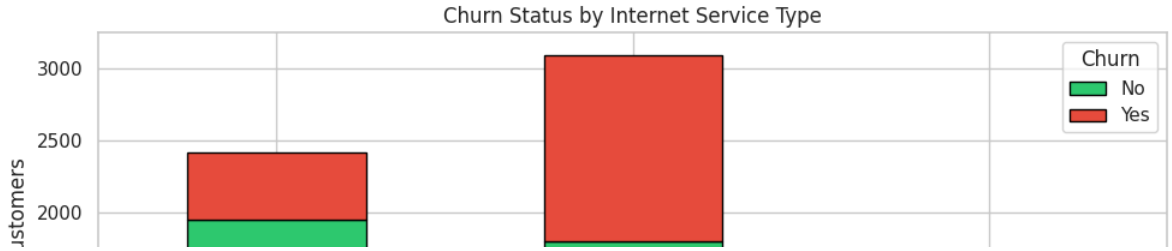
print(internet_churn)

internet_churn.plot(
    kind="bar",
    stacked=True,
    figsize=(10, 5),
    color=["#2ecc71", "#e74c3c"],
    edgecolor="black"
)

plt.title("Churn Status by Internet Service Type")
plt.xlabel("Internet Service Type")
plt.ylabel("Number of Customers")
```

```
plt.xticks(rotation=0)
plt.legend(title="Churn")
plt.tight_layout()
plt.show()
```

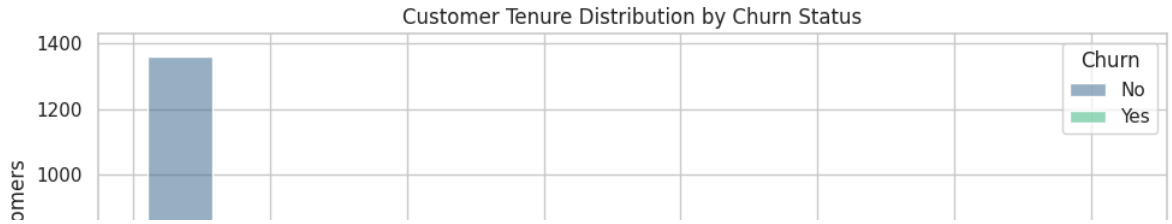
```
Churn              No   Yes
InternetService
DSL              1957   459
Fiber optic      1799  1297
No               1407   113
```




```
import matplotlib.pyplot as plt
import seaborn as sns

# Visualize the distribution of tenure (number of months with the company)
plt.figure(figsize=(10, 5))
sns.histplot(data=df, x="tenure", hue="Churn", multiple="stack", kde=True, p

plt.title("Customer Tenure Distribution by Churn Status")
plt.xlabel("Tenure (Months)")
plt.ylabel("Number of Customers")
plt.tight_layout()
plt.show()
```




```
# Bin the tenure into cohorts (groups of 12 months/1 year intervals)
bins = [0, 12, 24, 36, 48, 60, 72]
labels = ['0-12 Months', '13-24 Months', '25-36 Months', '37-48 Months', '49
df['tenure_group'] = pd.cut(df['tenure'], bins=bins, labels=labels)

# Calculate retention and churn rates per cohort
tenure_cohorts = (
    df.groupby('tenure_group', observed=False)
    .agg(
        total_customers=('customerID', 'count'),
        churned_customers=('Churn', lambda x: (x == 'Yes').sum())
    )
)

tenure_cohorts['retained_customers'] = tenure_cohorts['total_customers'] - t
tenure_cohorts['Retention Rate (%)'] = (tenure_cohorts['retained_customers']
tenure_cohorts['Churn Rate (%)'] = (tenure_cohorts['churned_customers'] / te

display(tenure_cohorts)

# Plot the retention rates by tenure group
```

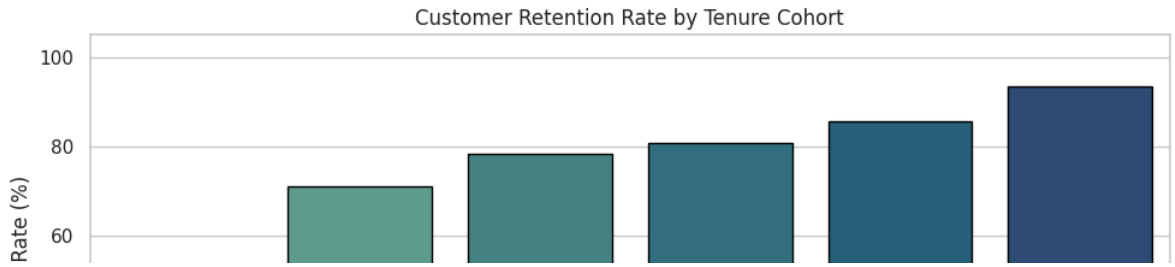
```
plt.figure(figsize=(10, 5))
sns.barplot(x=tenure_cohorts.index, y=tenure_cohorts['Retention Rate (%)'],
plt.title('Customer Retention Rate by Tenure Cohort')
plt.xlabel('Tenure Cohort')
plt.ylabel('Retention Rate (%)')
plt.ylim(0, 105)
plt.tight_layout()
plt.show()
```

tenure_group	total_customers	churned_customers	retained_customers	R
0-12 Months	2175	1037	1138	
13-24 Months	1024	294	730	
25-36 Months	832	180	652	
37-48 Months	762	145	617	
49-60 Months	832	120	712	
61-72 Months	1407	93	1314	

```
/tmp/ipykernel_667/1859209721.py:23: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be rer

  sns.barplot(x=tenure_cohorts.index, y=tenure_cohorts['Retention Rate
```



Next steps: Generate code with tenure_cohorts | New interactive sheet

```
# Identify new customers (tenure of 1 month or less) who are at a higher ris
new_customers = df[df["tenure"] <= 1]

# Filter for new customers with high monthly charges (above average)
avg_monthly = df["MonthlyCharges"].mean()
high_value_at_risk = new_customers[new_customers["MonthlyCharges"] > avg_mor

print("Total new customers (tenure <= 1 month):", len(new_customers))
print(f"Average monthly charge across all customers: ${avg_monthly:.2f}")
print("New high-value customers at immediate risk:", len(high_value_at_risk)
print("Churn count among these new customers:")
print(new_customers["Churn"].value_counts())
```

```
Total new customers (tenure <= 1 month): 613
Average monthly charge across all customers: $64.80
New high-value customers at immediate risk: 238
Churn count among these new customers:
```

```
Churn
Yes    380
No     233
Name: count, dtype: int64
```

```
fig, axes = plt.subplots(2, 2, figsize=(15, 10))

# 1. Distribution of Churn
sns.countplot(data=df, x="Churn", ax=axes[0, 0], palette="Set2", edgecolor='
axes[0, 0].set_title("Overall Churn Count")
axes[0, 0].set_xlabel("Churned")
axes[0, 0].set_ylabel("Number of Customers")

# 2. Churn Status by Internet Service Type
internet_churn.plot(kind="bar", stacked=True, ax=axes[0, 1], color=["#2ecc71
axes[0, 1].set_title("Churn Status by Internet Service Type")
axes[0, 1].set_xlabel("Internet Service Type")
axes[0, 1].set_ylabel("Number of Customers")
axes[0, 1].tick_params(axis="x", rotation=0)
axes[0, 1].legend(title="Churn")

# 3. Customer Retention Rate by Tenure Cohort
sns.barplot(x=tenure_cohorts.index, y=tenure_cohorts['Retention Rate (%)'],
axes[1, 0].set_title("Customer Retention Rate by Tenure Cohort")
axes[1, 0].set_xlabel("Tenure Cohort")
axes[1, 0].set_ylabel("Retention Rate (%)")
axes[1, 0].set_ylim(0, 105)
axes[1, 0].tick_params(axis="x", rotation=15)

# 4. Average Monthly Charges by Contract Type
sns.barplot(data=df, x="Contract", y="MonthlyCharges", ax=axes[1, 1], palett
axes[1, 1].set_title("Average Monthly Charges by Contract Type")
axes[1, 1].set_xlabel("Contract Type")
axes[1, 1].set_ylabel("Avg Monthly Charges ($)")

plt.suptitle("Telco Customer Churn & Retention Analytics Dashboard", fontsiz
plt.tight_layout()
plt.savefig("telco_churn_retention_dashboard.png", dpi=300, bbox_inches="tig
plt.show()
```

```
/tmp/ipykernel_667/3505345352.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be rer
```

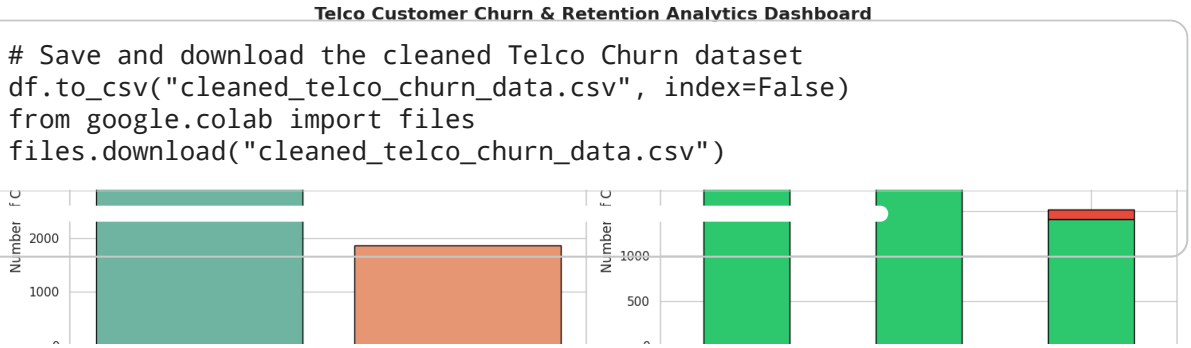
```
  sns.countplot(data=df, x="Churn", ax=axes[0, 0], palette="Set2", edg
```

```
# Export the updated cohort details to a CSV
tenure_cohorts.to_csv("tenure_cohort_retention.csv", index=True)

from google.colab import files

# Download the newly created outputs
files.download("telco_churn_retention_dashboard.png")
files.download("tenure_cohort_retention.csv")
```

```
  sns.barplot(data=df, x="Contract", y="MonthlyCharges", ax=axes[1, 1]
```




```
# Save and download the cleaned Telco Churn dataset
df.to_csv("cleaned_telco_churn_data.csv", index=False)
from google.colab import files
files.download("cleaned_telco_churn_data.csv")
```